Capture the Flag 8: Enroll a New StudentCapture the Flag 8: Enroll a New Student

🚩 Flag 0 — Create Your Feature Branch

Objective: Create a new feature branch before implementing student enrollment.

Create:

`feature/student-enrollment`

Screenshot:

```
C:\Users\Manes Christopher\Documents\mannes\CTF7\IT314AB_Artajo>git checkout -b feature/student-enrollment
Switched to a new branch 'feature/student-enrollment'

C:\Users\Manes Christopher\Documents\mannes\CTF7\IT314AB_Artajo>git branch
  feature/multi-screen-navigation
* feature/student-enrollment
```

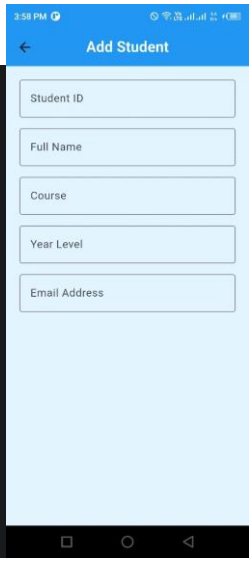
🚩 Flag 1 — Build the Add Student Form

Objective: Transform your existing Add Student Screen into a functional enrollment form.

Screenshot:

```dart
class _AddStudentScreenState extends State<AddStudentScreen> {
  final _idController = TextEditingController();
  final _nameController = TextEditingController();
  final _courseController = TextEditingController();
  final _yearController = TextEditingController();
  final _emailController = TextEditingController();

  @override
  void dispose() {
    _idController.dispose();
    _nameController.dispose();
    _courseController.dispose();
    _yearController.dispose();
    _emailController.dispose();
    super.dispose();
  }
```

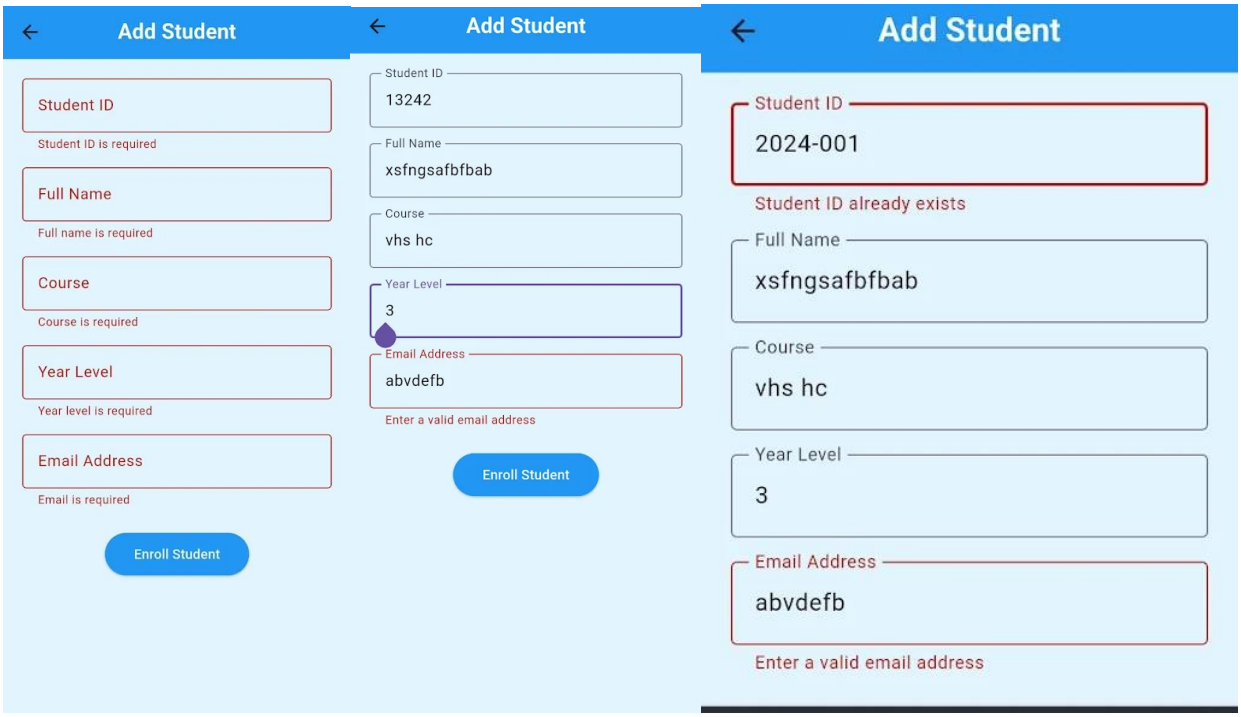


🚩 Flag 2 — Validate Student Information

Objective: Prevent incomplete, invalid, or duplicate student information from being submitted.

Screenshot:

```dart
validator: (value) {
  if (value == null || value.trim().isEmpty) {
    return 'Student ID is required';
  }
  final exists = students.any(
        (s) => s.studentId.toLowerCase() == value.trim().toLowerCase(),
  );
  if (exists) return 'Student ID already exists';
  return null;
},
```



🏴 Flag 3 — Enroll and Display a New Student

Objective: Add a successfully enrolled student to your existing Student Directory.

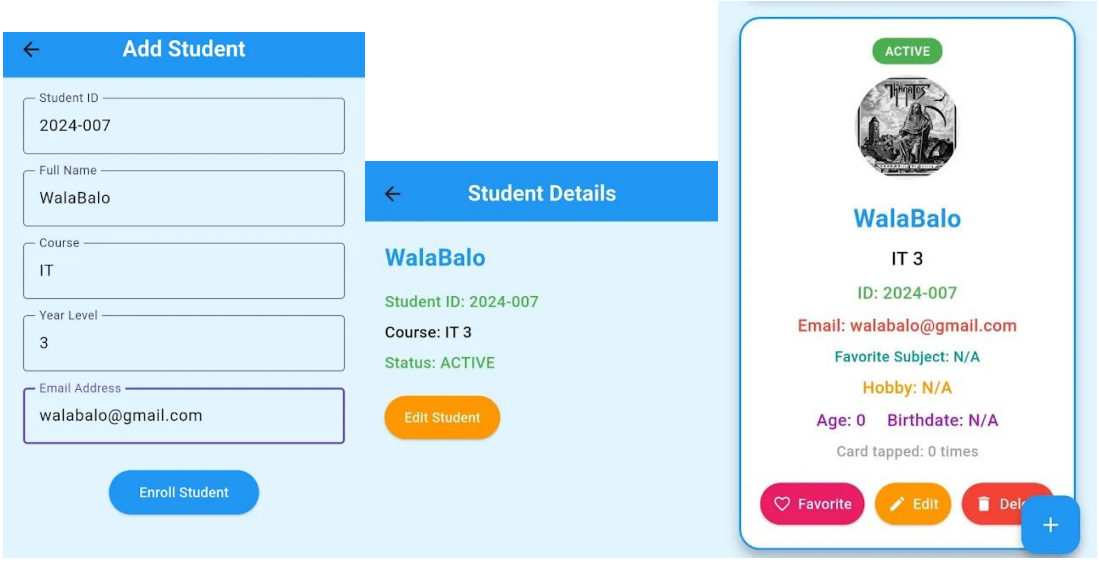
Screenshot:

```
onPressed: () {
  if (_formKey.currentState!.validate()) {
    final newStudent = Student(
      name: _nameController.text.trim(),
      course: '${_courseController.text.trim()} ${_yearController.text.trim()}',
      age: 0,
      hobby: 'N/A',
      image: 'assets/profile.jpg',
      birthdate: 'N/A',
      studentId: _idController.text.trim(),
      email: _emailController.text.trim(),
      favoriteSubject: 'N/A',
    ); Student

    students.add(newStudent);

    _idController.clear();
    _nameController.clear();
    _courseController.clear();
    _yearController.clear();
    _emailController.clear();

    Navigator.pop(context);
  }
},
style: ElevatedButton.styleFrom(
```



🚩 Flag 4 — Save Student Records Locally

Objective: Preserve student records even after the application is closed and restarted.

Screenshot:

```dart
// turn a map back into a Student (for loading)
factory Student.fromJson(Map<String, dynamic> json) => Student(
  name: json['name'],
  course: json['course'],
  age: json['age'],
  hobby: json['hobby'],
  image: json['image'],
  birthdate: json['birthdate'],
  studentId: json['studentId'],
  email: json['email'],
  favoriteSubject: json['favoriteSubject'],
  isFavorite: json['isFavorite'],
  isActive: json['isActive'],
);
```

```dart
// turn a Student into a map (por saving)
Map<String, dynamic> toJson() => {
  'name': name,
  'course': course,
  'age': age,
  'hobby': hobby,
  'image': image,
  'birthdate': birthdate,
  'studentId': studentId,
  'email': email,
  'favoriteSubject': favoriteSubject,
  'isFavorite': isFavorite,
  'isActive': isActive,
};
```

```dart
Future<void> main() async {
  WidgetsFlutterBinding.ensureInitialized();
  await loadStudents();
  students.sort((a, b) => a.name.compareTo(b.name));
  runApp(const MyApp());
}
```



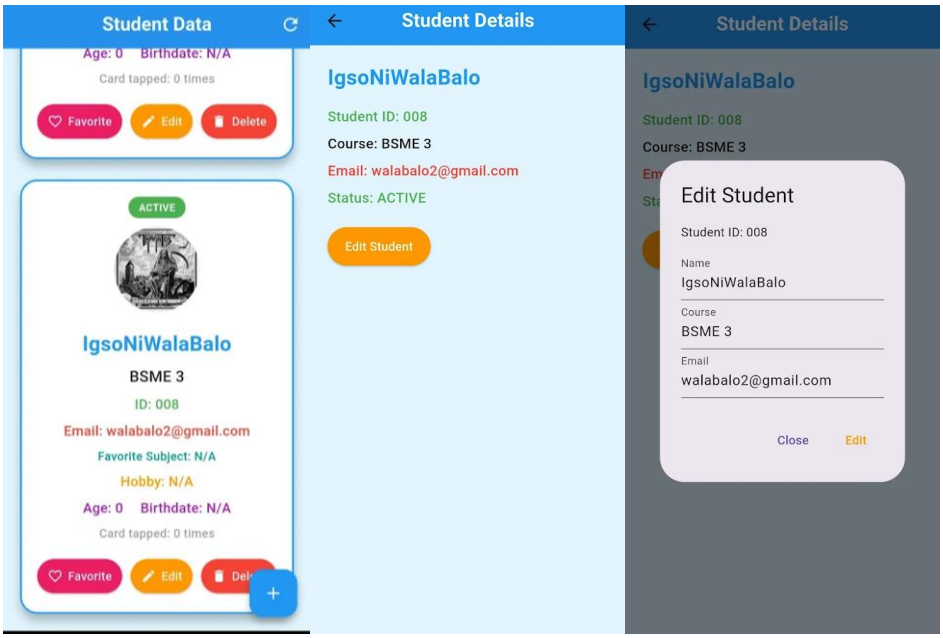

```dart
// save the whole list
Future<void> saveStudents() async {
  final prefs = await SharedPreferences.getInstance();
  final data = students.map((s) => jsonEncode(s.toJson())).toList();
  await prefs.setStringList('students', data);
}

// load the saved list, if nothing is saved yet keep the default list
Future<void> loadStudents() async {
  final prefs = await SharedPreferences.getInstance();
  final data = prefs.getStringList('students');
  if (data == null) return;
  students = data.map((s) => Student.fromJson(jsonDecode(s))).toList();
}
```

🏳 Flag 5 — Keep Existing Features Functional 🏳 Flag 5 — Keep Existing Features Functional

Objective: Ensure that enrollment works with the navigation and interactions implemented in CTF 7.

Screenshot:



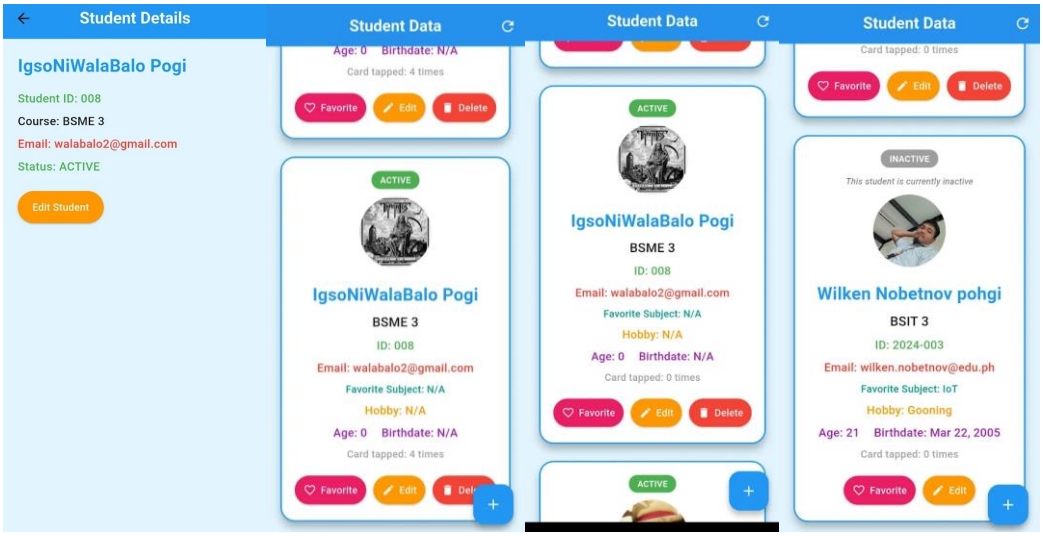


🏴 Flag 6 — Synchronize Changes with Local Data 🏴

Flag 6 — Synchronize Changes with Local Data

Objective: Ensure that creating, editing, and deleting students updates the locally saved records.

Screenshot:




```
students.add(newStudent);
saveStudents();

_idController.clear();
_nameController.clear();
_courseController.clear();
_yearController.clear();
_emailController.clear();

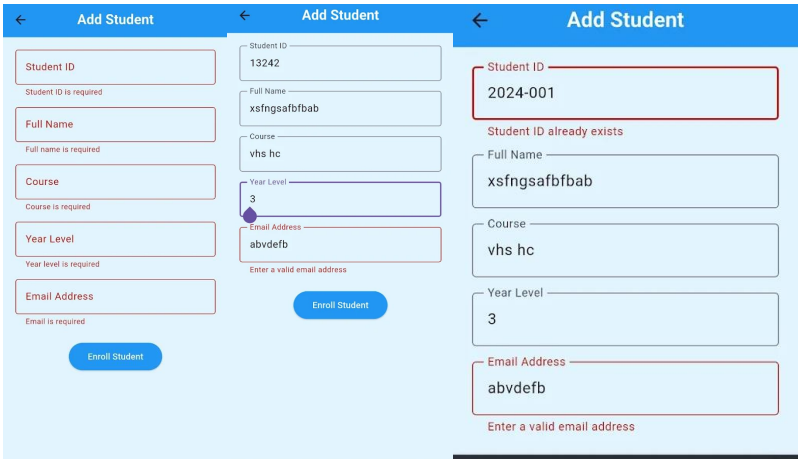
Navigator.pop(context);
}
```

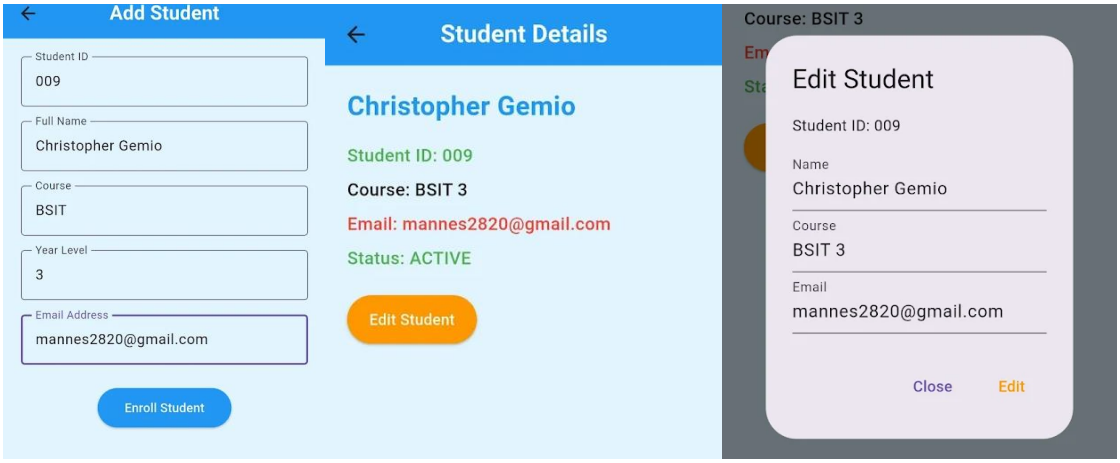
```
onPressed: () {
  setState(() {
    student.name = nameController.text.trim();
    student.course = courseController.text.trim();
    student.email = emailController.text.trim();
  });
  saveStudents();
  Navigator.pop(context);
  debugPrint('Edit confirmed for: ${student.name}');
},
child: const Text(
  'Edit',
  style: TextStyle(color: Colors.orange),
), Text
), TextButton
```

🚩 Flag 7 — Verify the Complete Enrollment Workflow

Objective: Demonstrate that the Student Directory supports enrollment, validation, navigation, editing, and local data persistence.

Screenshot:





Before The Restart



After The Restart (It changes the position Alphabetically)



🏴 Flag 8 — Save Your Progress 🏴 Flag 8 — Save Your Progress

Objective: Commit and push your completed Student Enrollment feature.

Use a proper Git commit message.

Branch:

`feature/student-enrollment`

Screenshot:

- Successful commit.
- Current branch.
- Terminal showing the commit.
- Successful push to the remote repository.

```
C:\Users\Manes Christopher\Documents\mannes\CTF7\IT314AB_Artajo>git branch
  feature/multi-screen-navigation
* feature/student-enrollment

C:\Users\Manes Christopher\Documents\mannes\CTF7\IT314AB_Artajo>git log --oneline -1
cf6fdce (HEAD -> feature/student-enrollment, origin/feature/student-enrollment) feat: add student enrollment with validation and local persistence

C:\Users\Manes Christopher\Documents\mannes\CTF7\IT314AB_Artajo>git status
On branch feature/student-enrollment
Your branch is up to date with 'origin/feature/student-enrollment'.

nothing to commit, working tree clean
```